Md Shamimul Islam

M.S. Student and Graduate Research Assistant, Computer Science, Florida Polytechnic University

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Research Summary

I work on AI for cybersecurity, computer vision, and LLM agents, building systems that are accurate, explainable, and practical to deploy. My M.S. thesis develops multi-agent LLM frameworks that detect network attacks and learn new rules from their own errors, now being extended to federated settings. I also fine-tune large and small language models for malware and intrusion detection, and build vision models for video-based safety monitoring and brain tumor segmentation. I am seeking a Ph.D. position starting Fall 2027.

Research Interests

AI for Cybersecurity	Network security and intrusion detection, malware detection, IoT security, trustworthy AI, federated learning
LLMs and Agentic AI	Large and small language models, retrieval-augmented generation, multi-agent systems
Computer Vision	Video understanding and human activity recognition, medical imaging, multimodal learning

Education

2025 – 2027 (expected) **M.S. in Computer Science**, Florida Polytechnic University, USA
CGPA: 3.77/4.00. Thesis: experience-augmented LLM/SLM frameworks for network intrusion detection. Advisors: Dr. Ayesha S. Dina and Dr. Luis G. Jaimes.
Coursework: Machine Learning and Practice (A), Scientific Computation and Programming (A), Advanced Algorithm Design and Analysis (A); in progress: Intelligent Systems, Thesis I.

2020 – 2022 **M.Sc. in Applied Statistics and Data Science**, Jahangirnagar University, Bangladesh
CGPA: 3.96/4.00, ranked 1st in batch. Thesis: Efficient Deep Learning Methods for Continuous Bangla Sign Language Recognition.
Coursework: Machine Learning for Data Science, Statistical Inference, Multivariate Analysis, Applied Regression Analysis, Time Series Analysis and Forecasting (all A+), Data Mining (A).

2015 – 2020 **B.Sc. in Computer Science and Engineering**, Manarat International University, Bangladesh
CGPA: 3.99/4.00, ranked 1st in batch. Thesis: Pose-based Multi-level Features Fusion for Exam Cheat Recognition Using CNN.
Coursework: Artificial Intelligence, Computer Vision and Robotics, Neural Networks and Fuzzy Systems, Digital Signal and Image Processing, Linear Algebra, Statistics and Probability (all A+).

Publications

Google Scholar (October 2026): 105 citations, h-index 4, i10-index 2. * Equal contribution.

Peer-reviewed conference papers

[1] J. Al-Mahmud*, **M. S. Islam***, and M. N. Hossain, "Video-based sexual harassment detection using fine-tuned video transformers," *IEEE 3rd International Conference on Computing, Applications and Systems* (*COMPAS*), University of Dhaka, Bangladesh, 2026. arXiv:2610.15354, code.

[2] **M. S. Islam**, L. G. Jaimes, and A. S. Dina, "MA-IDS: Multi-agent RAG framework for IoT network intrusion detection with an experience library," *IEEE International Conference on Systems, Man, and Cybernetics* (*SMC*), Bellevue, WA, USA, 2026. arXiv:2604.05458, code.

[3] S. Jannat, M. M. Hasan, **M. S. Islam**, and M. N. Hossain, "BaT-SAM: Introducing inductive bias into SAM via a mixture of convolutional experts for brain tumor segmentation," *12th IEEE International Women in Engineering* (*WIE*) *Conference on Electrical and Computer Engineering* (*WIECON-ECE*), 2026.

[4] **M. S. Islam**, A. J. M. A. Joha, N. Hossain, M. M. Hasan, and S. Abdullah, "Multi-view Bangla sign language recognition: A new word-level video dataset," *IEEE 6th International Conference on Activity and Behavior Computing* (*ABC*), Kyushu, Japan, 2024. doi:10.1109/ABC61795.2024.10652018.

[5] **M. S. Islam** and I. Elwarfalli, "Deep learning-powered ECG-based biometric authentication," *IEEE International Conference on Next-Generation Computing, IoT and Machine Learning* (*NCIM*), Dhaka, Bangladesh, 2023. doi:10.1109/NCIM59001.2023.10212845.

[6] N. H. M. Arafat, S. Ileas, Mujazid, and **M. S. Islam**, "BloSecR: A conceptual framework of identity documents' security on blockchain in refugee issues and conserving fundamental rights," *IEEE International Conference for Intelligent Technologies* (*CONIT*), Karnataka, India, 2022. doi:10.1109/CONIT55038.2022.9847731.

[7] **M. S. Islam**, M. M. Hasan, and S. Abdullah, "A deep spatio-temporal network for vision-based sexual harassment detection," *IEEE International Conference on Emerging Technology in Computing, Communication and Electronics* (*ETCCE*), 2021. doi:10.1109/ETCCE54784.2021.9689891.

[8] M. M. Hasan, **M. S. Islam**, and S. Abdullah, "Robust pose-based human fall detection using recurrent neural network," *IEEE International Conference on Robotics, Automation, Artificial-Intelligence and Internet-of-Things* (*RAAICON*), Dhaka, Bangladesh, 2019. doi:10.1109/RAAICON48939.2019.23.

Under review and in preparation

[9] **M. S. Islam** and A. S. Dina, "SERA-IDS: Structured experience retrieval-augmented intrusion detection with small language models," submitted to *IEEE International Conference on Communications* (*ICC*), 2027.

[10] **M. S. Islam**, A. H. Sohag, and Z. Subah, "PermissionGuard: Fine-tuned LLMs and BERT for Android permission-based malware detection," manuscript in preparation.

Datasets

[11] A. J. M. A. Joha, M. N. Hossain, **M. S. Islam**, S. Abdullah, and M. M. Hasan, *Multi-View Bangla Sign Language* (*MV-BSL*) *Dataset*, 2024. github.com/Mahedi-61/MV-BSDL.

[12] S. Abdullah, **M. S. Islam**, and M. M. Hasan, *Bangladeshi Vehicle License Plate Dataset*. github.com/Mahedi-61/Bangla_License_Plate_Dataset.

[13] **M. S. Islam** and M. M. Hasan, *Sexual Harassment Video Dataset* (*SHVD*). github.com/shamim-cs/Sexual-Harassment-Video-Dataset-SHVD-.

Research Projects

2025 – present **Experience-augmented LLM/SLM frameworks for intrusion detection**, M.S. thesis
Built MA-IDS (Multi-Agent Intrusion Detection System), where a Traffic Classification Agent and an Error Analysis Agent turn misclassifications into human-readable rules stored in a FAISS experience library (macro F1 89.75% on NF-BoT-IoT and 85.22% on NF-ToN-IoT, IEEE SMC 2026). Extended it into SERA-IDS (Structured Experience Retrieval-Augmented Intrusion Detection), which uses structured, confidence-gated rules so small local language models reach 69.5–87.7% macro F1 without fine-tuning. Now extending the framework to federated settings.

2025 – 2026 **Video transformers for sexual harassment detection**, IEEE COMPAS 2026
Extended the SHV dataset to 600 annotated clips and fine-tuned TimeSformer, VideoMAE and ViViT pretrained on Kinetics-400; TimeSformer reached 92.27% test accuracy.

2026 **Parameter-efficient fine-tuning of LLMs for intrusion detection**, course project, CAP5610
Compared QLoRA fine-tuning of LLaMA 3.1 8B with BERT, deep learning and classical ML baselines on the 10-class NF-UNSW-NB15-v3 benchmark. Code.

2026 – present **PermissionGuard: LLM-based Android malware detection**
Compared LoRA-tuned LLaMA 3.1-8B and BERT with classical ML models for permission-based malware detection, reaching up to 97.00% accuracy.

2024 – 2025 **AI career counselor and guidance platform**
LLaMA-powered platform for personalized IT career recommendations, built on an 8,000-sample dataset.

2024 **Phishing URL detection**
ML-based detection system using Logistic Regression, Random Forest, SVM and neural networks.

2022 – 2023 **Real-time Bangla sign language translation**
Transformer that translates pose-based sign language video representations into sentences; winner of AI for Bangla 2.0.

2022 **Bangla social network post analysis**
Transformer-based model for political sentiment and toxicity classification of Bangla social media posts.

2018 – 2019 **Automatic license plate recognition**
Real-time system on a custom Bangladeshi license plate dataset; published at IEEE RAAICON 2019.

Research and Teaching Experience

2025 – present **Graduate Assistant**, Department of Computer Science, Florida Polytechnic University
Graduate Research Assistant since Spring 2026; Graduate Teaching Assistant in Fall 2025.

2024 – 2025 **Lecturer**, Department of CSE, Manarat International University

2023 – 2024 **Lecturer**, Department of CSE, Asian University of Bangladesh
Convener, AUB Programming Club.

2020 – 2022 **Adjunct Faculty**, Department of CSE, Manarat International University

Courses taught: Artificial Intelligence, Machine Learning, Neural Networks, Object-Oriented Programming, Software Engineering.

Awards and Honors

2023 **Winner, AI for Bangla 2.0 Competition**
Led a team in the national competition run by the Government of Bangladesh, building an AI solution for Bangla sign language among 100+ teams.

2020 **Islamic Development Bank (IsDB) Masters Scholarship**
Competitive scholarship ($6,000) for graduate study.

2018 **University Merit Scholarship**, Manarat International University
Full tuition waiver for placing in the top 1% of undergraduate students.

2015 **Champion, Great Victory Day Essay Competition**, Manarat International University

Talks and Presentations

2026 **Invited Speaker, Engineering and Science Student Research Symposium**, Florida Polytechnic University
"Adaptive Reasoning for Network Traffic Security: A Multi-Agent RAG Architecture with Persistent Knowledge Induction."

2026 **Poster Presentation**, Florida Polytechnic University
"MA-IDS: Multi-Agent RAG Framework for IoT Network Intrusion Detection with an Experience Library."

2022 **Invited Speaker, Webinar on "A Publication Road Map for Scientific Papers"**, MIU CSE Club

Leadership and Service

2026 – present **Vice President, Muslim Students Association**, Florida Polytechnic University

2026 – present **Research and Workshop Coordinator, AI, Robotics, and Vision Research Club (AIRVRC)**, Florida Polytechnic University

2026 **Volunteer, Summer Camp 2026**, SALAM Academy Florida

2019 – 2020 **Elected President, MIU CSE Club**, Manarat International University
Organized MIU CSE Fest 2020, the intra-university programming contest and Code Camp.

2017 – present **Co-founder, Hasimukh (Smile)**, Rangpur, Bangladesh
Social welfare organization supporting underprivileged communities through education support and other giving.

2018 – 2019 **Competitions**
ICPC Bangladesh National Collaborative Programming Contest (2018); GDG Sonargaon Hackathon (2019).

Skills

Programming	Python, Java, C, R, SQL, PHP
ML and DL	PyTorch, TensorFlow, Hugging Face Transformers, PEFT/LoRA, scikit-learn, XGBoost
LLMs and agents	LangChain, FAISS, retrieval-augmented generation, multi-agent pipelines, local LLM/SLM deployment
Computer vision	OpenCV, YOLO, RetinaNet, Mask R-CNN, OpenPose, MediaPipe
Tools	Git, Linux, Google Cloud Platform, Google Colab, Kaggle, LaTeX
Languages	Bengali (native), English (fluent)

References

Available upon request.